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Assignment - CCNA - Routing and switching

Module -5 Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

Answer: c) Forwarding data packets between networks

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

Answer: d) Dynamically assigning IP addresses to devices

3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

Answer: b) Switch

4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

Answer: b) Bus

Section 2: True or False

1. True or False: A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

Answer: True

2. True or False: TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

Answer: False

3. True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Answer: True

Section 4: Practical

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

Answer – Done

Section 5

10. Discuss the Importance of Network Documentation in the Context of Building and Managing Networks

Answer -

Network documentation is the process of recording and maintaining detailed information about a network's design, configuration, devices, and operations. It serves as a reference for network administrators and IT teams when building, managing, troubleshooting, and upgrading a network.

Network documentation is important because it provides a clear understanding of how the network is structured and how different devices are connected. It helps administrators quickly identify and resolve network issues, reducing downtime and improving efficiency.

Proper documentation includes information such as network diagrams, IP address schemes, device configurations, hardware inventory, security settings, user access details, and maintenance records.

Importance of Network Documentation

1. Simplifies Troubleshooting

When a network problem occurs, documentation helps administrators locate the issue quickly and take corrective action.

2. Improves Network Management

It provides a complete overview of the network, making it easier to monitor, maintain, and manage network resources.

3. Supports Network Expansion

Documentation helps organizations plan upgrades and add new devices without affecting existing network operations.

4. Enhances Security

Maintaining records of devices, configurations, and access controls helps identify unauthorized changes and improve network security.

5. Reduces Downtime

Accurate documentation allows faster recovery from failures and minimizes interruptions to network services.

6. Assists New Administrators

New IT staff can understand the network structure and operations more easily through well-maintained documentation.

Conclusion

Network documentation is an essential part of building and managing networks. It improves troubleshooting, security, maintenance, and future planning while ensuring that network operations remain efficient and reliable.

1. Which of the following messages in the DHCP process are broadcasted? (Choose two)

Answer: B. Offer and C. Discover

2. Which command would you use to ensure that an ACL does not block web-based TCP traffic?

Answer: B. permit tcp any any eq 80

3. Explain Network Topologies

Answer -

Network topology refers to the physical or logical arrangement of devices in a network. It defines how computers, switches, routers, and other devices are connected and communicate with each other.

Types of Network Topologies

1. Bus Topology

All devices are connected to a single backbone cable. Data travels through the main cable to reach other devices.

2. Star Topology

All devices are connected to a central device such as a switch or hub. It is the most commonly used topology in modern networks.

3. Ring Topology

Each device is connected to two other devices, forming a circular path for data transmission.

4. Mesh Topology

Every device is connected to every other device. It provides high reliability because multiple communication paths exist.

5. Tree Topology

A combination of bus and star topologies used in large organizations.

4. Explain TCP/IP Networking Model

Answer -

The TCP/IP model is a set of communication protocols used for transmitting data over networks, including the Internet.

Layers of TCP/IP Model

1. Application Layer

Provides services to users and applications.

Examples:

- . HTTP**
- . FTP**
- . SMTP**
- . DNS**

2. Transport Layer

Responsible for end-to-end communication and data delivery.

Protocols:

- TCP
- UDP

3. Internet Layer

Handles logical addressing and routing of data packets.

Protocol:

- IP (Internet Protocol)

4. Network Access Layer

Controls how data is transmitted over the physical network.

Examples:

- Ethernet
- Wi-Fi

The TCP/IP model enables communication between devices connected to different networks.

5. Explain LAN and WAN Network

Answer -

LAN (Local Area Network)

A LAN is a network that connects computers and devices within a limited area such as a home, school, office, or building.

Characteristics of LAN

- Covers a small area
- High-speed communication
- Easy to manage

- **Lower cost**

Example

An office network connecting computers, printers, and servers.

WAN (Wide Area Network)

A WAN is a network that connects devices over large geographical areas such as cities, countries, or continents.

Characteristics of WAN

- **Covers a large area**
- **Connects multiple LANs**
- **Managed by service providers**
- **Higher communication cost**

Example

The Internet is the largest WAN.

6. Explain Operation of Switch

Answer -

A switch is a networking device used to connect multiple devices within a LAN. It operates at the Data Link Layer (Layer 2) of the OSI model.

When a device sends data, the switch examines the destination MAC address and forwards the data only to the intended device instead of broadcasting it to all devices.

Working of a Switch

- 1. Receives data from a device.**
- 2. Reads the destination MAC address.**
- 3. Checks its MAC address table.**
- 4. Forwards the data to the correct port.**
- 5. Updates the MAC table if necessary.**

Advantages

- Faster communication**
- Reduced network traffic**
- Better network performance**

7. Describe the Purpose and Functions of Various Network Devices

Answer -

Router

Connects different networks and forwards data packets between them using IP addresses.

Switch

Connects devices within a LAN and forwards data using MAC addresses.

Hub

Connects multiple devices and sends data to all connected devices.

Modem

Connects a network to the Internet Service Provider (ISP).

Firewall

Monitors and filters incoming and outgoing network traffic based on security rules.

Access Point

Provides wireless network connectivity to devices.

Repeater

Strengthens and regenerates weak network signals to extend network coverage.

Gateway

Acts as an entry and exit point between different networks.

8. Make a List of Appropriate Media, Cables, Ports, and Connectors Used to Connect Switches

Answer -

Media Type	Cable Used	Port	Connector
Copper Ethernet	UTP Cat5e/Cat6	Ethernet Port	RJ-45
Fiber Optic	Single Mode Fiber	SFP Port	LC Connector
Fiber Optic	Multi Mode Fiber	SFP Port	LC Connector
Console Connection	Console Cable	Console Port	RJ-45 / USB
High-Speed Connection	DAC Cable	SFP/SFP+ Port	SFP Connector

Common Ports on Switches

- **Ethernet Port**
- **Gigabit Ethernet Port**
- **SFP Port**
- **Console Port**
- **USB Port**

9. Define Network Devices and Hosts

Answer -

Network Devices

Network devices are hardware components that help establish, manage, and control communication within a network.

Examples:

- **Router**
- **Switch**
- **Hub**
- **Firewall**
- **Access Point**
- **Modem**

Their main purpose is to transmit, receive, and manage network traffic.

Hosts

A host is any device connected to a network that can send or receive data.

Examples:

- **Desktop computers**
- **Laptops**
- **Servers**
- **Smartphones**
- **Printers**
- **IP Cameras**

Each host is identified by an IP address and communicates with other devices on the network.

Module 8 – Network Access

1. Explain Switch

Answer -

A switch is a networking device used to connect multiple devices such as computers, printers, servers, and access points within a Local Area Network (LAN). It operates mainly at the Data Link Layer (Layer 2) of the OSI model and uses MAC addresses to forward data to the correct device.

Unlike a hub, a switch sends data only to the intended destination device, which improves network performance and reduces unnecessary traffic.

Functions of a Switch

- **Connects devices within a network**
- **Learns and stores MAC addresses in a MAC address table**
- **Forwards data to the correct destination**
- **Reduces network congestion**
- **Improves communication efficiency**

Advantages

- **Faster data transmission**
 - **Better security than a hub**
 - **Efficient use of network bandwidth**
-

2. Explain Switch Boot Sequence

Answer -

The switch boot sequence is the process followed by a Cisco switch when it is powered on.

Steps in the Switch Boot Sequence:-

Step 1: Power-On Self-Test (POST)

The switch checks its hardware components such as CPU, memory, and interfaces to ensure they are functioning properly.

Step 2: Load the Boot Loader

The boot loader program stored in ROM starts running and prepares the switch to load the operating system.

Step 3: Load Cisco IOS

The Cisco IOS (Internetwork Operating System) image is loaded from Flash memory into RAM.

Step 4: Load Startup Configuration

The switch searches for the startup configuration file stored in NVRAM and loads it.

Step 5: Switch Becomes Operational

The switch applies the configuration settings and starts normal operation.

Boot Sequence Flow

Power On → POST → Boot Loader → Load IOS → Load Startup Configuration → Normal Operation

3. Explain Three Methods to Access Switch Command Line Interface (CLI)

Answer -

The Command Line Interface (CLI) allows administrators to configure and manage a switch.

1. Console Access

Console access is the most common method used for initial switch configuration.

Requirements

- **Console cable**

- Computer or laptop
- Terminal software (PuTTY, Tera Term)

Advantages

- Direct access
 - Works even if network settings are not configured
-

2. Telnet Access

Telnet allows remote access to the switch over a network.

Features

- Uses TCP Port 23
- Allows remote management

Disadvantage

- Data is transmitted in plain text and is not secure
-

3. SSH (Secure Shell) Access

SSH provides secure remote access to the switch.

Features

- Encrypts communication
- Uses TCP Port 22
- Preferred over Telnet

Advantages

- Secure authentication
 - Protects login credentials and data
-

4. Explain and Configure the Cisco Internetwork Operating System (IOS)

Answer -

Cisco IOS (Internetwork Operating System) is the operating system used on Cisco routers and switches. It provides the commands and services required to configure, manage, and monitor network devices.

Functions of Cisco IOS

- Network configuration**
- Security management**
- Routing and switching operations**
- Device monitoring and troubleshooting**

Basic Cisco IOS Configuration

Enter Privileged Mode

Switch> enable

Enter Global Configuration Mode

Switch# configure terminal

Set Hostname

Switch(config)# hostname SW1

Configure Enable Password

SW1(config)# enable secret cisco123

Configure Console Password

```
SW1(config)# line console 0
```

```
SW1(config-line)# password cisco
```

```
SW1(config-line)# login
```

Save Configuration

```
SW1# copy running-config startup-config
```

This basic configuration allows administrators to secure and manage the switch.

5. Explain Switch Port

Answer -

A switch port is a physical interface on a switch used to connect network devices such as computers, printers, servers, routers, and other switches.

Each switch port provides a communication path between the connected device and the network.

Types of Switch Ports

1. Access Port

An access port belongs to a single VLAN and is typically used to connect end devices such as PCs and printers.

Example

A computer connected to VLAN 10 through an access port.

2. Trunk Port

A trunk port carries traffic from multiple VLANs between switches or between a switch and a router.

Example

A connection between two switches carrying VLAN 10, VLAN 20, and VLAN 30 traffic.

Functions of Switch Ports

- **Connect devices to the network**
- **Transmit and receive data**
- **Support VLAN communication**
- **Provide network access**

Common Port Types

- **Fast Ethernet (10/100 Mbps)**
- **Gigabit Ethernet (1 Gbps)**
- **10 Gigabit Ethernet**
- **SFP Ports for fiber connections**

Switch ports play a vital role in connecting devices and enabling communication within a network.

4-R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network

engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

Answer –

A. R1 and B. R2

3. enable secret [password] is hashed using the algorithm.

Answer: A. MD5

4. An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

Answer: D. Router 2.2.2.2 is a backup designated router.

5. Which command is used to view the neighbor discovery table on a PC?

Answer: C. netsh interface ipv6 show neighbor

6-What type of variable is being shown? Routers = [R1,R2,R3]

Answer: A. List

7. Identify the fields in an IPv4 header. (Choose three)

Answer: B. Time to Live , C. Source address and D. Destination address

Module 9 – Infrastructure Services

8. Host A and Host B sit in two different subnets. Which part of the APIC-EM Path Trace ACL Analysis tool analyzes ACL impact on packets?

Answer: D. Analysis of the impact of ACLs on the packets that would flow from Host A to B

9. Which IPv6 address is the equivalent of the IPv4 interface loopback address 127.0.0.1?

Answer: A. ::1

10. Which command is used to apply an ACL to an interface?

Answer: B. ip access-group

11. Which command and mode will successfully configure a hostname of R1 on a Cisco IOS router?

Answer: C. Router(config)# hostname R1

12. Which of the following reserved IPv4 addresses has binary 0s in all of the host bit positions?

Answer: D. Network address

Module 10 – Infrastructure Security and WAN Technologies

13- A Cisco Catalyst switch connects to what should be individual user PCs. Each port has the same port security configuration, configured as follows: interface range gigabit Ethernet 0/1 - 24 switchport mode access switchport port-security switchport port-security mac-address sticky Which of the following answers describe the result of the port security configuration created with these commands? (Choose two)

Answer: B. If a user connects a switch to the cable, prevents multiple devices from sending data through the port. And

C. Will allow any one device to connect to each port, and will save that device's MAC address into the startup-config.

14. What is the Administrative Distance of Internal EIGRP Routes?

Answer: B. 90

15. When a Subnet Mask is Presented in Binary, What do the Binary 1s Represent?

Answer: A. The network portion of an associated address

16. Which Switch Would STP Choose to Become the Root Bridge?

Answer: A. 32768: 11-22-33-44-55-66

17. Which Device is Used by the Service Provider to Provide WAN Services?

Answer: C. WAN switch

18. Cisco Router Acting as DHCP Server. Which Command Displays Assigned Addresses?

Answer: D. show ip dhcp bindings

19. Which Command Enables EIGRP Only on Interfaces with IP Addresses from 10.1.1.0 through 10.1.1.63?

Answer: A. network 10.1.1.0 0.0.0.63

20-R3 has a static route configured that points toward the service provider. What command could you use to have R3 advertise an OSPFv3 default route to the internal network, regardless of whether R3 had its default static route?

Answer: E. Have R3 use the command default-information originate always in OSPFv3 router configuration mode.

21. Which command is used to verify the interfaces that are being used as the outside interface and the inside interface?

Answer: F. show ip nat statistics

22. When using the "show EtherChannel summary" command, what does the "U" flag signify?

Answer: C. In use

23. Which command could you enter to encrypt passwords?

Answer: D. All of the above

24. Which command is used to identify the IPv4 addresses that will be in the DHCP pool?

Answer: A. network

25. ACL Processing on Router Interfaces (Choose Two)

Answer: A. Inbound ACLs will be processed before the routing table lookup occurs and

C. Outbound ACLs will be processed after the routing table lookup has occurred

26. OSPFv2 – Which Conditions Could Prevent Routers from Learning All Routes? (Choose Two)

Answer: A. An ACL could be blocking router advertisements

And

C. Any physical layer problem that would prevent two neighbouring routers from being able to ping each other's IPv4 addresses

27. Which Statements Describe Neighbor Discovery Functionality in IPv6? (Choose Two)

Answer: A. Determines the link layer address of a neighbor and D. Queries for duplicate addresses

28. Which IPv6 Prefix Will the Typical Enterprise Network Receive From the Service Provider?

Answer: E. /48

29. How Should a Switch Be Configured So That It Could Be Accessed Remotely?

Answer: C. Configure a gateway for the switch

30-Refer to the exhibit. A network technician is asked to design a small network with redundancy. The exhibit represents this design, with all hosts configured in the same VLAN. What conclusions can be made about this design?

Answer: C. The router will not accept the addressing scheme.